



Computer Graphics mesh

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Today's Topics: mesh

- **Mesh**
 - Concepts
 - Mesh simplification
 - Mesh subdivision

Concepts—Background of mesh



形体表示: Two main type of modelers(模型)

- Solid-based (基于实体的表达方法)
 - usually seen in the area of *Computer-Aided-Design* (CAD), often emphasize modeling tools correspond to actual machining processes, such as cutting, drilling(钻孔), etc.
(强调对应 实际工业生产中的加工过程, 如切割, 钻孔等)
 - Surface-based (基于表面的表达方法)
 - 曲面造型, 没有实体的概念, 但易于显示。
 - Bezier、B样条曲线曲面适用于表达工业造型中较光滑曲线曲面
 - do not have a built-in concept of solidity(不能有效表达真实世界中的物体, 如茶壶、兔子模型);
 - 图形学中广泛使用三角网格来表达三维模型。
- ✓ 三角网格 (离散表面表示方法) 已成为最为广泛使用的模型表达方法。



Concepts

- Mesh Description ((三角) 网格描述)
 - A list of faces(一系列面片) $F = (f_1, f_2, \dots, f_n)$
 - Each face is a **triangle**
 - A list of vertices (一系列顶点) $V = (v_1, v_2, \dots, v_n)$
 - Each face in F is a list of indices in V (V 中顶点的序列组)
e.g.
 $f_1 \rightarrow (v_1, v_2, v_3), f_2 \rightarrow (v_4, v_5, v_6),$
 $f_3 \rightarrow (v_7, v_8, v_9), \dots$

计算机中，我们都是用很多这样的三角形来表示三维复杂的模型，而表示这样的三角网格模型，我们要有两个**要素**。一个是点表，一个是面表。



Some Samples

- Examples of triangle mesh models, a bull, a dragon and a head model. We draw triangles over bull model, and show the other two models by shading(明暗处理).

- 模型“牛”上显示了三角网格的结构。
- 模型“龙”和“人头”则是利用三角网格进行绘制的结果。



Why mesh representation?



- Computer generated 3D models and captured data have different representations
(计算机产生的三维模型和实际获取的数据的表示模式不同)。
 - We need a unified representation in graphics
 - visual accuracy (视觉精度) and speed should be acceptable
 - Due to fast development of graphics hardware, we could rasterize(光栅化) and render triangles very quickly nowadays(能快速地进行光栅化和三角网格的渲染操作).

Source of 3D data(三维数据的来源)



- **Models could be generated by(模型产生途径)**
 - Directly typing in the geometric file(直接在几何文件中输入)
 - Writing code to create such data: *procedure modeling*(程序建模)
 - Using modeling software such as 3ds/max, maya(运用建模软件创建)
 - Capturing(获取) a real model using a 3D scanner(3D扫描仪获取真实模型)
 - Reconstruction(重构) from one or more photographs (照片), called *photogrammetry*(照相测量法, 图形学中经典难题)
 - Combination of techniques above Others...



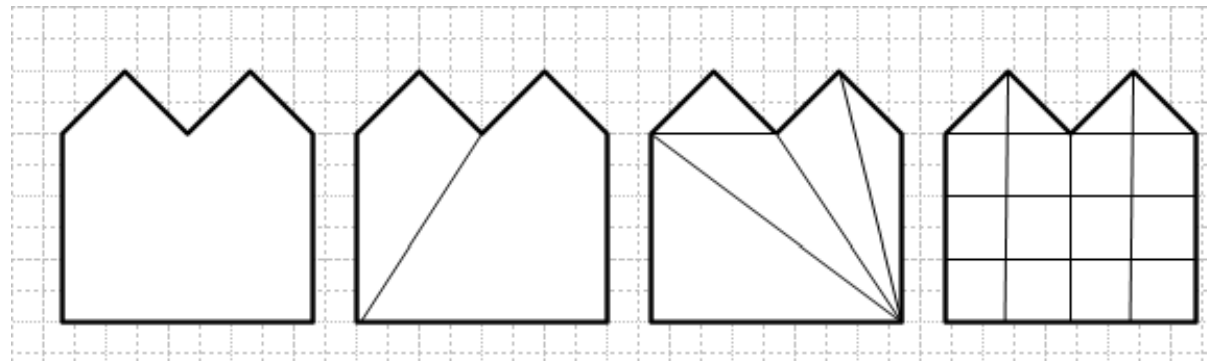
Mesh Tessellation(网格化)

- 给你一个图形(平面图形或空间的表面), 怎样把它变成网格的表示。
- Tessellation in 2D (以2D 网格化为例)
 - We introduce 2D Tessellation because it's useful in 3D mesh processing.
 - Polygons can arrive in(表示成)many different forms, and may have to be split into(分割)more tractable primitives (更易于处理的图元), such as convex(凸面的)polygons, triangles or quads(四边形), this process is called tessellation(网格化). If the polygon is split into triangles, this process is called triangulation(三角化).



Mesh Tessellation(网格化)

- Various types of tessellation.



- The leftmost polygon is not tessellated. The next is partitioned into convex regions, the next is triangulated, and the rightmost is uniformly meshed(均匀化的分割).



Simplification(简化)

□ 真实感的实时绘制问题变的日益突出，因为做实时绘制，而开始了模型简化，如电影和游戏需求。

- What is Mesh Simplification?

- approximating a given input mesh with a less complex but geometrically faithful representation

(用一个相对简单但保持足够几何细节的网格模型来近似表示一个给定的网格)

(减少一个已有网格的面片数量，同时仍能表达原三维模型)



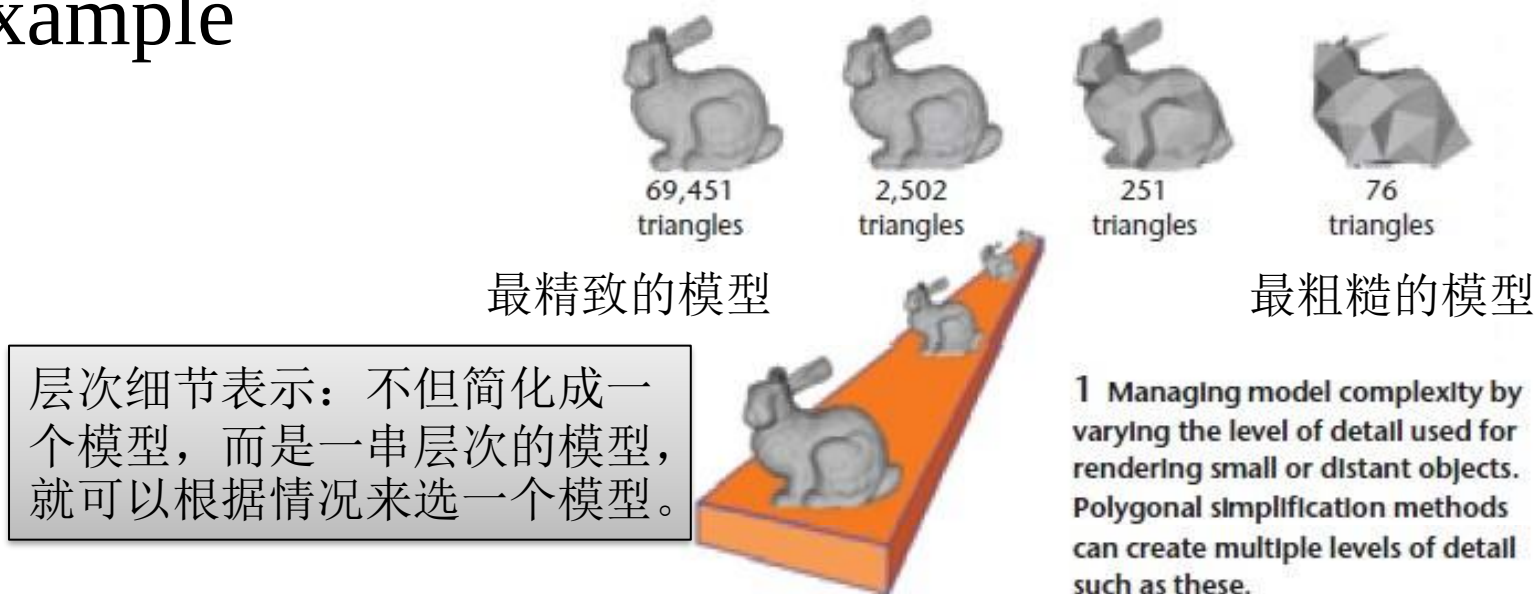
Simplification(简化)

- **Why do we need Mesh Simplification?**
 - **remove redundant geometry (移除几何冗余)**
 - E.g. a flat region(平坦区域) with many small, coplanar(共面的) triangles. Merging these triangles to large polygons could decrease model's complexity.
 - **reduce model size**
 - reduce the size to store or transmit it(降低存储或传输的规模).
 - **improve run-time performance(运行性能)**
 - simplification can be essential to efficient rendering(简化对高效的渲染具有重要作用)—图形的实时性非常重要, 比如动画游戏。网格简化是实时显示的一种重要方法。如, 当一个模型离你足够远的时候, 并不需要用那么精致的模型, 可以简化它, 用粗糙的模型来渲染。



层次细节网格简化技术

- generating **levels of detail** (LOD, **层次细节**) of objects in a scene. Presenting(呈现) distant(远的) objects with a lower LOD and near object with a higher LOD.
- An example



- 层次细节显示简化技术：在不影响画面视觉效果的情况下，通过逐次简化景物的表面细节来减少场景的几何复杂性，提高绘制算法的效率。



层次细节显示简化技术的基本原理

- 该技术对一个原始多面体模型建立几个不同逼近程度的几何模型。
 - 从近处观察物体时，采用精细的模型，
 - 从远处观察物体时，采用较粗糙的模型。
 - 当视点连续变化时，在两个不同层次的模型之间存在一个明显的跳跃，有必要在相邻层次的模型之间形成光滑的视觉过渡，即：几何形状过渡(geomorphs)。

层次细节简化的研究方向

总之，层次细节简化技术的研究主要集中在：

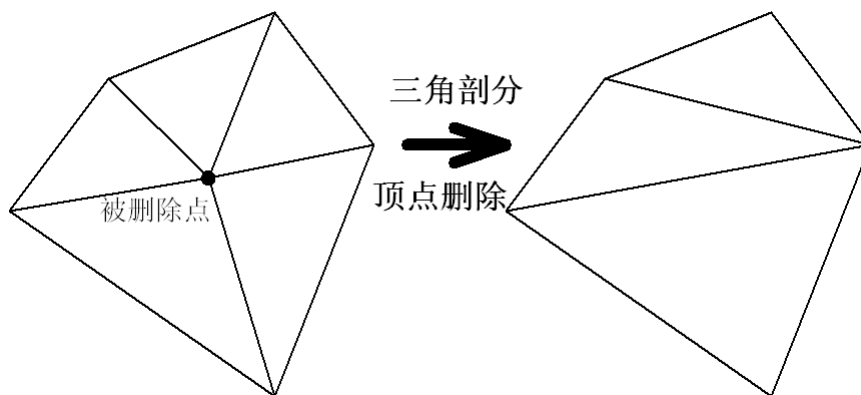
- 建立不同层次细节的模型。对任意给定的复杂多边形网格，由精细至粗糙建立一模型序列。
- 建立相邻层次的多边形网格之间的几何形状过渡。



层次细节模型化简的基本操作

□ 原始网格模型的不同层次细节模型的建立过程中，存在如下三种不同的基本化简操作：

1. **顶点删除操作**：删除网格中的一个顶点，然后对它的相邻三角形形成的空洞作三角剖分，保持网格的拓扑一致性。

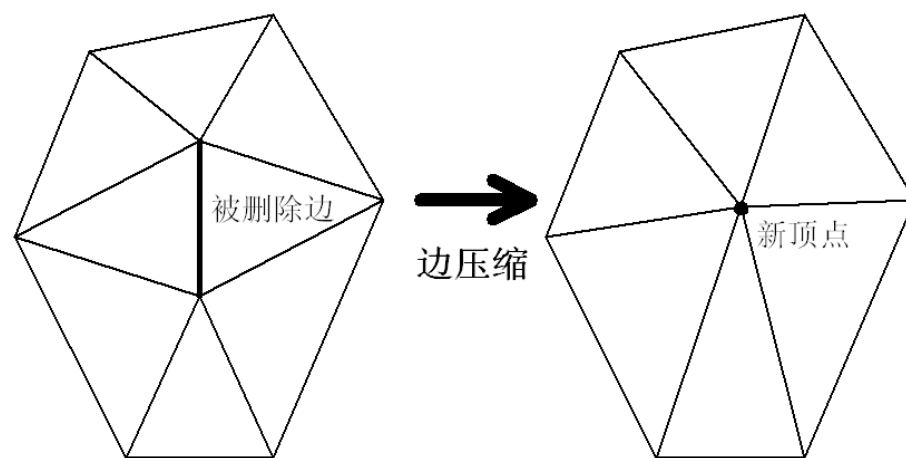




网格化简的基本操作

- 三种不同的基本化简操作：

2. **边压缩操作**：网格上的一条边压缩为一个顶点，与该边相邻的两个三角形的退化。

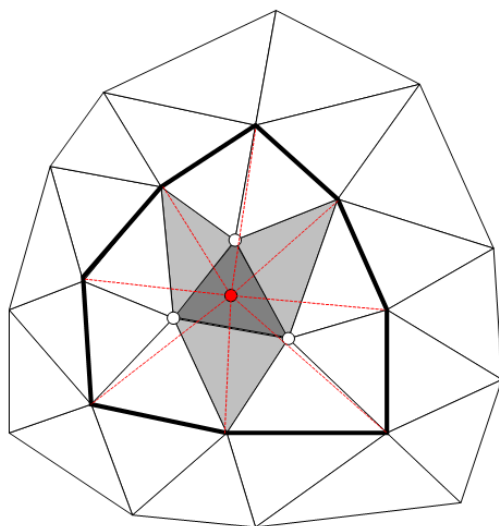




网格化简的基本操作

- 三种不同的基本化简操作：

3. **面片收缩操作**：网格上的一个面片收缩为一个顶点，该三角形本身和与其相邻的三个三角形都退化。



Simplification

网格模型处理过程中的一些重要概念



- Topology (拓扑结构)
 - refers to the connected polygonal mesh's structure
(连通多边形网格的结构)
 - genus (亏格)
 - the number of holes in the mesh surface.(网格表面孔洞的数目)
 - For example, a sphere(球体) and a cube(立方体) have a genus of zero(亏格数目为零), while a torus(环) and a coffee cup have a genus of one.
 - *local topology* of a face, edge, or vertex(面片, 边或顶点的局部拓扑)
 - refers to the connectivity of that feature's neighborhood.
(指相邻元素的局部连接关系, 如三角形有三条边, 一般有三个三角形与它相邻。
又如每条边有两个三角形与它相邻。)



Simplification(简化)

- Mechanism(技术方法): how to simplify?
 - Nearly every simplification technique in the literature uses some variation or combination of four basic polygon removal mechanisms below:
(四种基本多边形移除技巧的变种或组合)
 - sampling,
 - adaptive subdivision (自适应细分),
 - decimation (去除),
 - and vertex merging (顶点合并) .



Mechanism

- Sampling(采样)
 - Sampling algorithms sample the initial model's geometry, with points on the model's surface.
(通过选取模型表面的点简单地对模型进行几何取样)
These methods are among the more elaborate (复杂的) and difficult to code approaches.
 - They may have trouble to sample the high-frequency features accurately(难于精确获取高频特征, 比如说丰富表面细节). These algorithms usually work best on smooth surfaces with no sharp corners(尖角).

Feature sensitive Re-meshing(特征敏感的再分片)



Mechanism

- Adaptive subdivision(自适应细分)
 - Adaptive subdivision algorithms find a simple *base mesh* (基底网格) that can be recursively(递归的) subdivided(细分) to closely approximate the initial model(寻找一个可以递归细分 逼近最初模型的 基网格). This approach works best when the base model(基底模型) is easily found(基底模型易于获取).
 - Adaptive subdivision methods preserve the surface topology(表面拓扑细节), which may limit their capacity for drastic(大的) simplification. (对模型进行大规模简化的能力不足).



Mechanism

- Decimation(去除)
 - Decimation techniques iteratively(迭代的) remove **vertices** or **faces** from the mesh, re-triangulating the resulting hole after each step(并三角化每步移除后留下的孔洞).
 - These algorithms are relatively simple to code and can be very fast. Most use strictly local changes that tend to preserve the genus. (大部分应用限制局部变化并且易于保持亏格)

These algorithms are good at removing redundant geometry (几何冗余) such as coplanar(共面的) polygons (尤其适用于处理像共面多边形这种冗余的几何) .



Mechanism

- Vertex-merging(顶点合并)
 - 顶点合并一般将模型两个或多个顶点合并为一个顶点。
 - Vertex merging is a simple and easy-to-code mechanism, but algorithms use various techniques to determine which vertices to merge in what order.
 - Edge-collapse algorithms(边坍塌算法), which always merge two vertices sharing an edge, tend to preserve local topology, but algorithms permitting general vertex-merge operations can modify topology. (该算法通常会保持局部拓扑, 但也允许修改拓扑)

Subdivision(细分)

网格细分是和网格简化相反的(opposite)操作



- Subdivision
 - the publication of the papers by Catmull and Clark, Doo and Sabin in 1978 marked the beginning of subdivision for surface modeling(标志着曲面细分造型的开始). Now we can regularly see subdivision used in movie production.

(细分算法广泛应用于电影等行业)

- a loose(简略)description of subdivision
 - Given a original mesh, generate a smoother one by a sequence of successive refinement(通过网格加细(连续的细化)产生更光滑的效果)





Subdivision(细分)

- Example of Subdivision in 1D

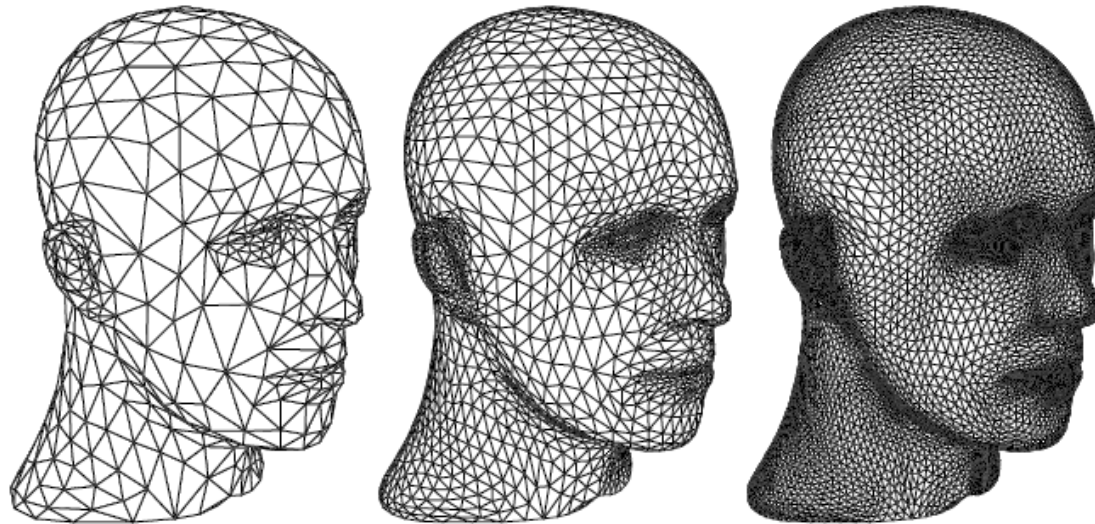


- On the left 4 points connected with straight line segments.(最左面的 4 个点由直线段相连)
- To the right of it a refined(加细的) version: 3 new points have been inserted between the old points.
(每条边外面加一个点, 连起来, 替换原来的边)
- After two more steps of subdivision the curve starts to become rather smooth.



Subdivision(细分)

- Another Example of Subdivision (人头模型也一样)



Example of subdivision, showing 3 successive levels of refinement.
(三个连续的细分的结果). On the left an initial triangular mesh. Each triangle is split into 4 according to a particular subdivision rule (middle). On the right the mesh is subdivided in this fashion once again.
(再次进行前面的细分得到最右边的结果, 使得网格不断的变得更光滑)



Subdivision(细分)

- Subdivision process (细分过程)
 - the subdivision of surfaces can be thought of as (被看成) a **two-phase process**.
 - The first phase, called the *refinement phase*(细化阶段), creates new vertices and reconnects to create new, smaller triangles.
(并与先前顶点相连产生新的、更小的三角形)
 - The second phase, called the *smoothing phase*(平滑阶段), computes the new positions for some or all vertices.(计算新顶点的位置)
 - The details of these two phases determine a subdivision scheme(这两步的细节决定了不同的细分方案). In the first phase, a triangle can be split(分割) in different ways; in the second phase, the positions for vertices could be interpolated differently.(新顶点的位置可以不同的方式插值产生)

Subdivision Scheme(细分方案)



- Subdivision Schemes
 - we will introduce 2 subdivision scheme
 - Loop Subdivision(Loop 细分)
 - $\sqrt{3}$ Subdivision



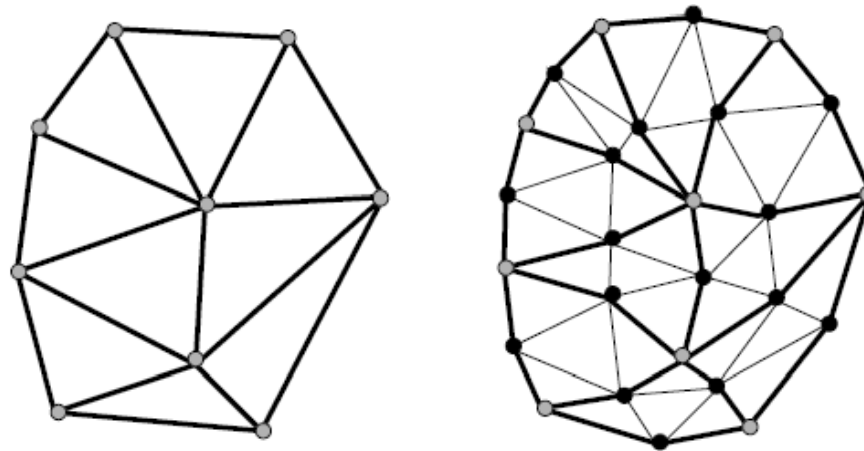
Loop Subdivision

- Loop Subdivision
 - Loop subdivision scheme was the first subdivision scheme for triangles.(第一个基于三角网格的细分方案)
 - 两个步骤：第一步是增加顶点；第二步是对顶点位置进行调整，使网络更加平滑。
 - It updates each existing vertex and creates a new vertex for each edge, then each triangle is subdivided(分割) into 4 new triangles. So after n subdivision steps, a triangle will be subdivided into $4n$ triangles.



Loop Subdivision

- Loop Subdivision



Example of Loop Scheme. New vertices are shown as black dots. a new vertex is added on each edge, and new vertices are reconnected to form 4 new triangles, replacing each triangle of the mesh.



Loop Subdivision

- Denotation(相关定义) of subdivision rule
 - Focus on an existing vertex p^k , where k is the number of current subdivision steps(细分步数). This means that p^0 is the vertex of the control mesh. In general, subdivision processes
$$p^0 \rightarrow p^1 \rightarrow p^2 \rightarrow p^3 \dots$$
 - If the valence(度) of p^k is n , then p^k has n neighbor vertices, denote the neighbors are $p_i^k, i \in \{0, 1, \dots, n-1\}$



Loop Subdivision

- Subdivision rule

- Below gives the subdivision rules of Loop's scheme (下面给出Loop细分方案的规则). The first formula is the rule for updating an existing vertex from p^k into p^{k+1} , and the second formula is for creating a new vertex p_i^{k+1} , on the edge between p^k and p_i^k .

(当前顶点和周围一圈邻居顶点的加权)

$$p^{k+1} = (1 - n\beta)p^k + \beta(p_0^k + \dots + p_{n-1}^k)$$

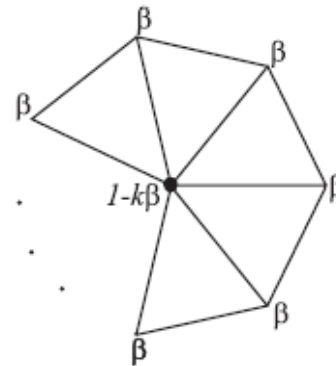
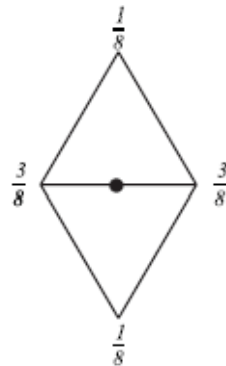
$$p_i^{k+1} = \frac{3p^k + 3p_i^k + p_{i-1}^k + p_{i+1}^k}{8}, i = 0, \dots, n-1$$

(创建一个新的点时，要考虑这个点的左右上下4个点)



Loop Subdivision

- Subdivision Rule



- This picture gives the weight for interpolating a vertex (left for adding a new vertex on an edge, right for updating an existing vertex)(这张图给出了插入顶点的规则模板 (左边是在边上加入一个新顶点 , 右边是对顶点的更新调整)
- β is the function of n as $\frac{1}{n}(5/8 - (\frac{3}{8} + \frac{1}{4} \cos \frac{2\pi}{n})^2)$



$\sqrt{3}$ - Subdivision

- $\sqrt{3}$ - Subdivision

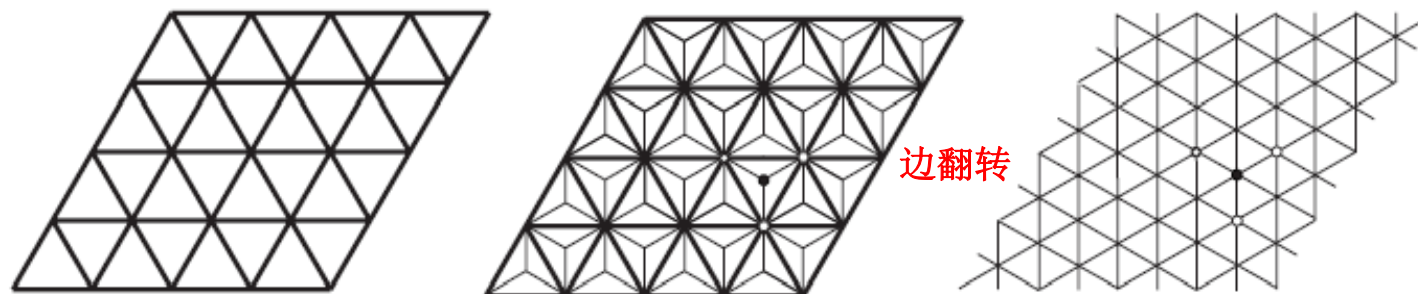
- Loop's scheme splits each triangle into four new ones, and so creates triangles at a rate of $4nm$ (以 $4nm$ 的比率创建新的三角形), where m is the number of triangles in control mesh, and n is the number of subdivision step.
- Different from Loop's scheme, Kobbelt's $\sqrt{3}$ scheme **creates only 3 new triangles per-step**.
- It creates a new vertex (called *mid-vertex*) **in the middle of each triangle**, instead of a new vertex per-edge (而不是在每个边上创建一个顶点).



$\sqrt{3}$ - Subdivision

- $\sqrt{3}$ - Subdivision

- To get more uniformly shaped triangles(更均匀分布的三角形), each old edge is flipped(翻转) so that it connects two neighboring mid-vertices, instead of connecting two neighboring existing vertices.



- this figure shows the process of $\sqrt{3}$ subdivision
- 原顶点与重心之间的边保留，原顶点与顶点之间的边翻转为新产生的重心与重心之间的边，减少了一些特别尖锐的三角形。
 - ✓ 经过一次该细分，三角形个数变成3倍，即一个变3个。
 - 边翻转并没有改变三角形的个数，只是改变了它的拓扑结构。

$\sqrt{3}$ - Subdivision



- Subdivision rule

- As shown in the formula in the first line, where p_m denotes the mid-vertex, computed as the average of the triangle vertices: p_a, p_b, p_c

Each of the existing vertices, p^k , are updated using the formula in the second line, where p_i^k denotes the neighbors of p^k . And as in Loop's scheme, n is the valence of p^k , and k is the subdivision step.

$$p_m^{k+1} = (p_a^k + p_b^k + p_c^k) / 3$$

$$p^{k+1} = (1 - n\beta) p^k + \beta \sum_{i=0}^{n-1} p_i^k$$



$\sqrt{3}$ - Subdivision

- Subdivision rule

- Again, β is a function of valence n , and the following choice of $\beta(n)$ generates a surface, where the continuity is at least C1) (1阶参数连续性) .

$$\beta(n) = \frac{4 - 2\cos(2\pi/n)}{9n}$$

- 如果 β 采用该函数, 已证明至少可以产生一个C1连续 (参数连续) 的曲面。



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第7讲课后作业:

- 1) What is Mesh Simplification? Why do we need Mesh Simplification?
- 2) What is Subdivision(细分)? 描述Subdivision process (细分过程) .
- 3) 大概介绍Loop Subdivision(Loop 细分)与 $\sqrt{3}$ Subdivision.